

ISEC2077
Security Auditing & Control Systems
The Machine Stops

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1. Why do you think I assigned this, specifically for a Cyber Security class?

This story, on its surface, is about a society that relies on, worships, and lives inside of one big machine. More than that, it critiques our dependence on tools in the advent of the Industrial Revolution, the limitation that those tools place upon our true knowledge, and how far this sense of “brain drain” can be pushed. Deeper than that, and closer to the point of the question: As technology grows more advanced and ubiquitous, its inner workings have become increasingly obscured to the average user. This story shows a world where people worship the machine and pray for its progress, yet because of blind faith in the machine, nobody knows how to maintain or repair it. In the real world, we have a duality of those who see the machine as infallible, but many who feel an active distrust out of misguided reasons. In one of Shawn’s classes, we watched an [interview](#) Sheriff Michael Bouchard, who proudly admitted his ignorance about his own device while viewing technical experts as essentially 'elitist wizards', another with an obscure and unfair power. These attitudes are increasingly common, and in cybersecurity, we have to persuade them to update their firmware. Whether you’re a professional involved in maintaining a system, or a user who relies on the product of its inner workings, you need to know what you’re working with, and both can fall victim to this form of ignorance. When professionals operate with uncritical faith in their tools, they lose the ability to identify anomalies, question false positives, or adapt their systems creatively to novel threats. In other ways, the story highlights the pitfalls of neglecting the core tenets of cybersecurity, from its “one big machine” creating a single point of failure, and human vulnerability vectors leading to a lack of backups, maintenance, or patching, creating a major incident without a recovery plan.

2. Do you see any relevance to your lives, focusing on how technology has changed in your lifetime and how you see people interact technology?

In my lifetime, technology has evolved from a tool we used to an environment we inhabit. We have migrated our lives to a digital platform, prizing the convenience of immediate information over the struggle to acquire it. The people of the Machine would certainly celebrate our “progress.” As our platforms and services consolidate, their inner workings become more of an impenetrable “black box” that very few are knowledgeable on, and this has only been exacerbated by the rise of artificial intelligence that can only function at a black box level of complexity and scale. “*The Machine Stops*” during Amazon Web Services outages. We have traded our innate knowledge of navigation for the constant guidance of GPS. A book that I recently read and enjoyed, [The Shallows by Nicholas Carr](#), argues that this trend is as much of a neurological shift as it is a social one. He argues that the internet is physically rewiring our brains for fragmented attention, systematically trading deep, linear thought for shallow and utilitarian information processing. It’s a [cognitive trade-off](#) where we gain immense processing power and connectivity at the potential cost of our capacity for deep focus and contemplation. Friedrich Nietzsche, upon receiving one of the first typewriters while nearly blind, found that the convenience of the machine altered his writing style to be more telegraphic, showing an example of how tools we use to express thought can reshape the thought itself. We’re constantly negotiating what we gain and what we lose with each new technological “progress,” just as the citizens of the Machine did, until they had nothing left to lose but the reality of progress itself.

Additional Thoughts & Comments :

Favorite passage?

"The machine may progress.

The machine may progress.

*The machine may progress,
eternally."*

This aphorism evolves in contextual meaning to some pretty great effect. At first, it's a hopeful phrase that wishes for more useful services from the machine. As time goes on, it becomes a theological chant by those who have come to totally rely on it. This becomes symbolic of the machine's initial purpose as a tool, and the evolution of its ubiquitous and omnipresent functionalities leaving its users without agency and completely controlled by it.

"We're all just prisoners here, of our own device"

- Hotel California, The Eagles

Alongside our common use of the word 'device', it's also used to describe projects and schemes. To be "left to your devices" means to be given time to do as you please. In the song Hotel California, the lyrics paint the world as a place where our 'devices' turn against us, in both a literal and metaphorical sense. The 'device' we build for the pleasure and conveniences of our 'devices' has become a mechanism of entrapment. By serving the progress of the machine, we forget how to live without it. By automating all lived experience, we lose sight of the purpose of the experience.

Does your point of view impact how the story reads?

Science fiction is one of my favorite modes of artistic expression. Through it, we can see different envisionments of reality and the possibilities of what our world could be, what it could have been, and what it needs to avoid. We may return to an old science fiction story and find a story packed with "predictions", but these ideas aren't happenstance or coincidence, but rather a reading of the times they were written in, and the pervading ideas and attitudes that evoked the inspiration to create them. Fans of fiction, fantasy, and games engage in "the magic circle", the idea that there exists a boundary between a fictional world and the real world. This boundary is what allows you to drop your realistic pretenses to become immersed in a "virtual" world. The boundary is also quite porous, where your frameworks of the virtual and the real can synthesize to create new understanding.

So, as a fan of science fiction and other forms of literature, I believe my perspective reading this story was to establish frameworks through which the characters are interacting, and see where my own knowledge of dystopian sci-fi writing can fill in the blanks. The frameworks my perspective establishes attempt to see the inner-workings of the world and how its depiction of it serves the real-world purpose of the author's messaging.

Would it have read the same before you took these classes?

My initial thoughts see the story as a warning of societal and philosophical dangers through loss of agency, isolation, and blind faith in progress. To be quite honest, that's still how I see it, but now I have a new set of analytical tools that provides me with layers of specificity and technicality to how the story played out, and why. Learning about the CIA triad, redundancy, points of failure, recovery plans, threat assessment, patching, and security awareness training, among others, has greatly enhanced how I read the story. These cybersecurity subjects are a massive benefit for my critical eye, because they allow me to more critically assess the storyline with a greater level of detail to ask the right questions with the 5W's. A benefit to my literary analysis, the magic circle allows me to test out my cybersecurity knowledge in a playful field, and it has done so with many other pieces of media since the start of this course.

Closing Thoughts

"I am beginning to think that it is true that the Machine did away with the necessity for human effort, and that we are now reaping the reward. We have lost the sense of space and the sense of touch; we have forgotten the very feel of our own bodies."

We are now witnessing the same elimination of human effort in the artistic and intellectual realms with the rise of large language models claiming to be "artificial intelligence". The tangible struggle of artistic endeavor is what creates its irreplaceable value. When one looks at artwork generated without human effort, the inconsistencies noticed over time create a natural feeling of revulsion called the 'uncanny valley.' Given the structural and technical limitations of these models, this feeling is inescapable, and I believe that greater integration will most likely lead to greater rejection. In cybersecurity, this isn't really up for debate. We are already seeing the consequences of this rush to automate core functions. Executives are flocking to vibecoded architectures, i.e. systems generated by AI that feel right but lack a secure, logical foundation. Meanwhile, threat actors gain an asymmetric advantage simply by asking the same models how to exploit the same code that they have generated.

It's a house of cards. How long can this last when the primary drivers of the AI industry, Nvidia, OpenAI, and Oracle rely on a potentially fraudulent circular economy of investment propping up an unsustainable model? We're rapidly approaching model collapse, where the only way forward is oceans of AI-generated data are used to train new AI, creating a feedback loop of escalating hallucinations and degraded performance. The perpetual promise that "this is the worst it will ever be" is beginning to sound less like a reassurance and more like the chant from *The Machine Stops*, worshipping the progress of a machine that we no longer understand and cannot realistically maintain.